

Data Collection and Preprocessing Phase

Date	07 Oct 2025
Team ID	Xxxxxx
Project Title	Predicting Plant Growth Stages with Environmental and Management Data Using Power BI
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The dataset used for this project is sourced from Kaggle – Plant Growth Data Classification . It contains various environmental and soil parameters such as temperature, humidity, sunlight hours, soil nutrients, and plant growth stage labels . The goal of this dataset is to classify plants based on their growth conditions and identify the factors that most influence growth.
Data Cleaning	During the cleaning process, missing values were identified and handled using mean or median imputation (depending on the feature type). Duplicate records were detected and removed to maintain data integrity. Outliers in numeric attributes such as temperature or sunlight hours were reviewed and adjusted using interquartile range (IQR) methods to ensure balanced representation.

Data Transformation	Power Query (or equivalent Python data transformation techniques) was used for filtering irrelevant rows , sorting data by growth stage, and pivoting tables to compare multiple conditions. Calculated columns such as average humidity index and growth ratio were created to improve data understanding and modeling efficiency.
Data Type Conversion	The dataset contained mixed data types (e.g., numeric values stored as text). These were corrected — numerical attributes (like temperature, humidity, and light hours) were converted to float, while categorical variables (like plant type or growth stage) were encoded as category or numerical labels for model compatibility.
Column Splitting and Merging	Columns with combined information (e.g., “Date and Time”) were split into separate columns for detailed analysis, while closely related features like “Nitrogen”, “Phosphorus”, and “Potassium” were merged into a composite nutrient index column to simplify model input.
Data Modeling	Relationships between variables were analyzed using correlation matrices and visualizations . Key measures such as average growth rate , mean sunlight hours , and humidity impact score were created. The processed dataset was then used to train and test various machine learning classification models .
Save Processed Data	The cleaned and transformed dataset was saved in .csv format as processed_plant_growth_data.csv . This file will be used for further model training, evaluation, and visualization to ensure reproducibility of results.